

Fitting Different Yield Curves to US Treasuries Data

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1 Introduction

This document details the mathematical implementation and empirical evaluation of the **Nelson-Siegel-Svensson (NSS)** and dynamic **Diebold-Li** term structure models implemented in Python. The objective of this project is to model U.S. Treasury yield dynamics using parameters calibration to fit and predict bond yields.

2 The Models

2.1 Nelson-Siegel-Svensson (NSS) Model

The Nelson-Siegel-Svensson (NSS) [1] model extends the original three-factor Nelson-Siegel formulation by introducing a second curvature component. This addition enables the framework to fit complex, non-monotonic yield curve shapes—such as dual humps, troughs, or inverted S-curves—that frequently emerge during shifting macroeconomic regimes.

For a given observation date t and time-to-maturity horizon $\tau > 0$, the zero-coupon yield $y_t(\tau)$ is expressed as a linear combination of four latent factor state variables $\beta_t = [\beta_{1t}, \beta_{2t}, \beta_{3t}, \beta_{4t}]^T$ weighted by non-linear loading functions of two decay parameters (λ_1, λ_2)

$$y_t(\tau) = \beta_{1t} + \beta_{2t} \left(\frac{1 - e^{-\lambda_1 \tau}}{\lambda_1 \tau} \right) + \beta_{3t} \left(\frac{1 - e^{-\lambda_1 \tau}}{\lambda_1 \tau} - e^{-\lambda_1 \tau} \right) + \beta_{4t} \left(\frac{1 - e^{-\lambda_2 \tau}}{\lambda_2 \tau} - e^{-\lambda_2 \tau} \right)$$

Taking the limit as $\tau \rightarrow 0$, the curvature factor loadings vanish and we are left with

$$\lim_{\tau \rightarrow 0^+} y_t(\tau) = \beta_{1t} + \beta_{2t} = y_{t, \text{shortest term}}$$

Intuitively, the shorter the maturity, the more bond yield converges to the baseline rate β_{1t} and the spread of current monetary policy β_{2t} . Conveniently, since the model is a linear combination of β_t , one can apply Ordinary Least Squares (OLS) to estimate the factor state vector. As for the decay rates, one can either choose two fix values and use them to fit the yield model, or proceed with two grid searches and then apply OLS. This paper will proceed with the latter.

1. **Decay Grid Initialization:** Define a bounded search grid for decay parameters $(\lambda_1, \lambda_2) \in (0, \bar{\lambda}] \times (0, \bar{\lambda}]$ subject to the identification constraint $\lambda_1 \neq \lambda_2$.

2. **Design Matrix Construction:** For a candidate pair (λ_1, λ_2) and maturity vector $\boldsymbol{\tau} = [\tau_1, \tau_2, \dots, \tau_N]^T$, construct the $N \times 4$ basis matrix $\mathbf{X}(\lambda_1, \lambda_2)$:

$$\mathbf{X} = \begin{bmatrix} 1, & \frac{1 - e^{-\lambda_1 \tau_i}}{\lambda_1 \tau_i}, & \frac{1 - e^{-\lambda_1 \tau_i}}{\lambda_1 \tau_i} - e^{-\lambda_1 \tau_i}, & \frac{1 - e^{-\lambda_2 \tau_i}}{\lambda_2 \tau_i} - e^{-\lambda_2 \tau_i} \end{bmatrix}$$

3. **Conditional OLS Extraction:** Evaluate the optimal factor state vector $\hat{\boldsymbol{\beta}}_t$ analytically using Ordinary Least Squares:

$$\hat{\boldsymbol{\beta}}_t(\lambda_1, \lambda_2) = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}_t$$

4. **Global SSE Minimization:** Select the decay pair $(\lambda_1^*, \lambda_2^*)$ that minimizes the cross-sectional Sum of Squared Errors (SSE):

$$(\lambda_1^*, \lambda_2^*) = \arg \min_{\lambda_1, \lambda_2} \left\| \mathbf{y}_t - \mathbf{X}(\lambda_1, \lambda_2) \hat{\boldsymbol{\beta}}_t(\lambda_1, \lambda_2) \right\|_2^2$$

2.2 Diebold-Li Model

While the static Nelson-Siegel-Svensson framework optimizes parameters independently across daily cross-sections, the dynamic **Diebold-Li model** [2] reinterprets the three-factor Nelson-Siegel specification as a time-series framework for latent macroeconomic factors. By fixing the decay parameter $\lambda = \bar{\lambda}$ across all observation dates, the factor loadings become deterministic and static, allowing the latent factor trajectories $\boldsymbol{\beta}_t = [\beta_{1t}, \beta_{2t}, \beta_{3t}]^T$ to be modeled and forecasted autoregressively.

For observation date t and maturity τ , the yield equation is expressed as:

$$y_t(\tau) = \beta_{1t} + \beta_{2t} \left(\frac{1 - e^{-\bar{\lambda} \tau}}{\bar{\lambda} \tau} \right) + \beta_{3t} \left(\frac{1 - e^{-\bar{\lambda} \tau}}{\bar{\lambda} \tau} - e^{-\bar{\lambda} \tau} \right)$$

where the factors keep their economic interpretations. In their original paper, Diebold & Li chose $\bar{\lambda} = 0.0609$. However, this paper will proceed with the same method as has been done with the NSS model.

1. **Decay Grid Initialization:** Define a bounded search grid for candidate decay parameters $\lambda \in (0, \bar{\lambda}]$.
2. **Design Matrix Construction:** For a candidate parameter λ and maturity vector $\boldsymbol{\tau} = [\tau_1, \tau_2, \dots, \tau_N]^T$, construct the static $N \times 3$ basis matrix $\mathbf{X}(\lambda)$:

$$\mathbf{X} = \begin{bmatrix} 1, & \frac{1 - e^{-\lambda \tau_i}}{\lambda \tau_i}, & \frac{1 - e^{-\lambda \tau_i}}{\lambda \tau_i} - e^{-\lambda \tau_i} \end{bmatrix}$$

3. **Vectorized Panel OLS Extraction:** Compute the full latent factor matrix $\hat{\boldsymbol{\beta}}_t(\lambda) \in \mathbb{R}^{T \times 3}$ across all T observation dates simultaneously:

$$\hat{\boldsymbol{\beta}}_t(\lambda) = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}_t$$

4. **Panel SSE Minimization:** Select the optimal fixed decay parameter $\bar{\lambda}^*$ that minimizes the cross-sectional Sum of Squared Errors across the entire panel $\mathbf{Y} \in \mathbb{R}^{T \times N}$:

$$\bar{\lambda}^* = \arg \min_{\lambda} \left\| \mathbf{y}_t - \mathbf{X}(\lambda) \hat{\beta}_t(\lambda) \right\|_2^2$$

3 Some Output

For static curve fitting, the paper chose US Treasury yields on 09/01/2026

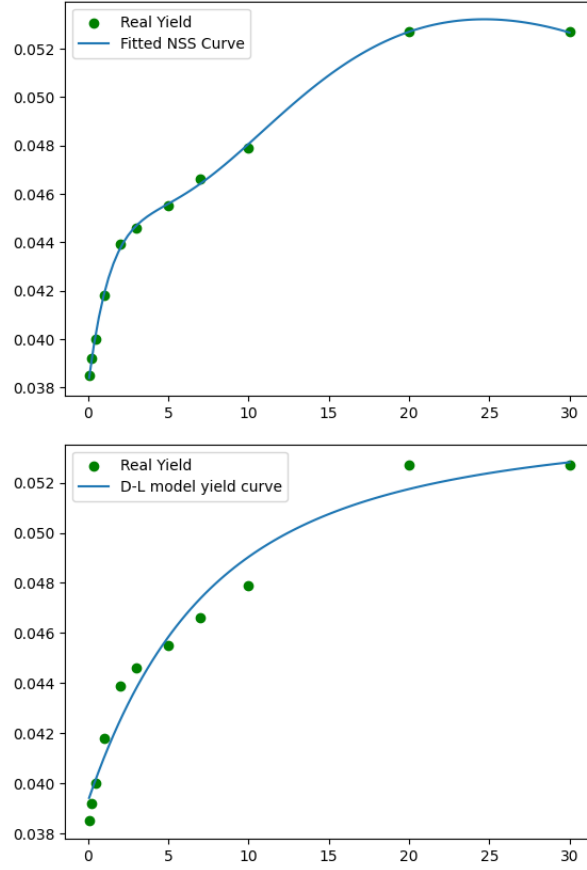


Figure 1: NSS yield curve vs DL yield curve. $SSE_{NSS} = 1.642 \times 10^{-7}$
vs $SSE_{DL} = 7.334 \times 10^{-6}$

The NSS model fits the data better, due to it having more factor to account for the sudden changes in yield data.

References

- [1] Lars E.O Svensson. *Estimating and Interpreting Forward Interest Rates: Sweden 1992 - 1994*. Working Paper 4871. National Bureau of Economic Research, Sept. 1994. DOI: 10.3386/w4871. URL: <http://www.nber.org/papers/w4871>.
- [2] Francis X. Diebold and Canlin Li. “Forecasting the term structure of government bond yields”. In: *Journal of Econometrics* 130.2 (2006), pp. 337–364. DOI: 10.1016/j.jeconom.2005.03.005.